

# Shaan Patel

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## OBJECTIVE

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Ambitious third-year Computer Engineering student specializing in computing hardware, VLSI design, and emerging architectures. Seeking a Fall 2026 internship in computer architecture, embedded systems, or hardware design to apply strong programming skills and an end-to-end approach to creating complex systems.

## EDUCATION

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### Georgia Institute of Technology

Atlanta, GA

*Bachelor of Science in Computer Engineering; GPA: 3.59*

*August 2023 – December 2026*

- Relevant Coursework: Data Structures and Algorithms, Embedded Systems Design, Advanced Microarchitecture Design
- Leadership: Captain of GT Ramblin' Raas, GTXR Executive Board, SiliconJackets Digital Design Team Lead

## TECHNICAL SKILLS

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**Programming:** C++, Verilog, C, VHDL, RISC-V, SystemVerilog, OpenGL, Python, Java, JavaScript, HTML/CSS, SQL, LaTeX

**Platforms:** Linux (Ubuntu, Slackware, Debian), Red Hat (OpenStack, CloudForms), OpenVMS, Solaris, Windows, MacOS

**Hardware:** FPGA, ESP32, IC Breadboard Circuits, Oscilloscope, Arduino, Raspberry Pi, ARM MBED, LaunchPad, 3D Printing

**Frameworks:** React, SFML, CUDA, PyTorch, AWS (S3, DynamoDB, EC2), Express, MongoDB, Pandas, Node.js, Kubernetes

**Software:** KiCAD, Altium, Cadence, Intel Altera Quartus II, LTspice, NI LabVIEW, gRPC, TCP/UDP Tunnels, Blender, Autodesk Fusion 360/AutoCAD, SolidWorks, Onshape, Visual Studio Code, GitHub, Wireshark, Adobe Creative Suite, Inkscape, Slack

**Communication:** Design proposals, technical reports, instruction manuals, presentations, email correspondence, training materials

## EXPERIENCE

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### AeroCore-V - Hardware-Accelerated SoC for Autonomous UAVs

Atlanta, GA

*SiliconJackets Project Lead*

*August 2025 – Present*

- Designed a RISC-V SoC in SystemVerilog with custom ALU extensions for single-cycle PID control math, tailored for low-latency flight adjustments on a DE10 validated with Intel Quartus Prime
- Implemented a two-level cache system with an LRU replacement policy optimized for high-frequency I2C/SPI GPS and IMU data streams from an ESP32 to reduce memory stalls by 62% compared to a single-level cache design
- Developed a virtual memory system with a protected frame table to isolate flight-critical control routines from telemetry code, ensuring system resilience against page faults
- Orchestrated a priority-based thread scheduler to manage UAV flight loops and telemetry tasks, achieving a 15% latency reduction validated via a networked OpenGL Digital Twin physics simulation

### Georgia Tech HIVE Makerspace

Atlanta, GA

*Student Leader and Volunteer*

*August 2024 – Present*

- Instruct ~40 students per week on advanced design and fabrication workflows, including 3D CAD/slicing, circuit analysis with oscilloscopes, laser cutter operation, PCB design and manufacturing (taking a blank FR4 sheet to an operational PCB)
- Deployed a multi-device print manager to automate jobs on a FIFO basis, increasing the operational uptime of the free-to-use 12-unit Bambu Lab 3D print farm for the Georgia Tech student body
- Facilitate workshops and created training resources that contributed to a 20% increase in student participation
- Architecting a full-stack inventory management system using an AWS RDS backend to track materials in the cloud with S3 and DynamoDB to enable a student electronics rental program, aiming to reduce annual student costs

### High-Performance Heterogeneous Computing Infrastructure

Ashburn, VA

*BlendFarm Project Leader*

*June 2022 – July 2025*

- Assembled two 2.5kW parallel computing systems from the component level, integrating twelve 12GB GDDR6 GPUs and modifying voltage curves to bypass LHR limiters, achieving a 73% hash rate improvement
- Overcame thermal throttling by designing and CNC-fabricating monolithic aluminum water blocks with Solidworks to cool a 6-GPU array, reducing core temperatures by over 15°C under sustained 100% load
- Scripted a distributed render farm using C++ and SFML to coordinate large ray tracing frame rendering across a network
- Optimized task allocation using TCP/UDP socket tunnels and a custom jitter buffer implementation to ensure reliable data synchronization between the master node and rendering clients for faster computer graphics generation

### Academy of Science International Research

Ashburn, VA

*Research Collaborator*

*November 2022 – July 2024*

- Synthesized a saline-soluble polymer as a novel alternative to currently used petrochemical-based plastics as a means of preventing marine pollution
- Modeled and fabricated 3D-generated molds using filament and resin printers for synthesizing hydrogels through incubation
- Worked with a tensile tester to measure Young's modulus and SEM electron microscopy to analyze sample composition
- Led collaboration efforts with South Korean research teams for two years to formulate samples and complete testing
- Collected over 150 trials of data and was awarded the Yale Outstanding Project award at RSEF in Spring of 2023